

Counit Edge-Case Counterexamples for Two Ideal Questions

Candidate negative answers to Questions 5.3 and 5.4 of arXiv:2111.13247

Automated mathematical discovery run `fa_banach_001`, lane 15

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Status. Candidate full counterexamples to the questions as written, likely valid, requiring expert review. The construction uses the permitted zero-ideal edge case.

Abstract

Questions 5.3 and 5.4 of Anderson-Sackaney have negative answers as stated. Let $\mathbb{G}$ be a noncoamenable compact quantum group and take the contractive convolution idempotent to be its universal counit $\epsilon_{\mathbb{G}}^u$. Its multiplier is the identity, so the associated compact quasi-subgroup is all of $L^\infty(\mathbb{G})$ and both ideals in the questions are zero. The zero ideal has the constant-zero bounded right approximate identity. On the other hand, the universal counit does not belong to the reduced measure algebra, since that would give a reduced counit and coamenability. The compact dual of the free group $\mathbb{F}_2$ gives a concrete example. Proper/nonzero variants are not settled.

1 The two source questions

The source asks whether a brai for $J^1(N)$ forces the idempotent state ω_N to belong to $M^r(\mathbb{G})$, and more generally whether the preannihilator of the range of every contractive-idempotent multiplier has a brai exactly when $\mathbb{G}$ is coamenable [1].

2 Counterexample

Theorem 1. *Questions 5.3 and 5.4 of [1] are both false as written.*

Proof. Let $\Gamma = \mathbb{F}_2$ and let $\mathbb{G} = \widehat{\Gamma}$ be its compact dual quantum group. Thus

$$C^u(\mathbb{G}) = C^*(\Gamma), \quad C^r(\mathbb{G}) = C_r^*(\Gamma).$$

The compact quantum group $\mathbb{G}$ is not coamenable because Γ is not amenable [2].

Let $\epsilon_{\mathbb{G}}^u \in M^u(\mathbb{G}) = C^u(\mathbb{G})^*$ be the universal counit. It is a unital character and hence a norm-one state. It is also the convolution unit, so

$$\epsilon_{\mathbb{G}}^u * \epsilon_{\mathbb{G}}^u = \epsilon_{\mathbb{G}}^u.$$

It is therefore a contractive idempotent of precisely the kind allowed in Question 5.4. The counit identity gives

$$M_{\epsilon_{\mathbb{G}}^u}^l = \text{id}_{L^\infty(\mathbb{G})},$$

an identity also recorded in the source's multiplier preliminaries.

We have made partial progress towards identifying when the left ideals $J^1(N)$ associated with a compact quasi-subgroup admit a brai. While we have a complete characterization in terms of the condition $J^u(N) = \overline{J^1(N)}^{wk*}$ (Theorem 4.14), our characterization in terms of coamenability of $\mathbb{G}$ (Corollary 4.16) requires what is essentially a coamenability type condition on N . This leaves us with the following question.

Question 5.3. *Given a CQG and compact quasi-subgroup N , if $J^1(N)$ has a brai, then do we have $\omega_N \in M^r(\mathbb{G})$?*

Successfully answering the above question means we can say $\mathbb{G}$ is coamenable if and only if $J^1(N)$ admits a brai.

We have also characterized coamenability of $\mathbb{G}$ in terms the existence of brais on the associated left ideals of a very small class of TROs associated with a contractive idempotent (Theorem 4.25 and Corollary 4.26). Namely, if we set $X = M_\omega^l(L^\infty(\mathbb{G}))$ where $\omega \in M^u(\mathbb{G})$ is a contractive idempotent, we require $Gr(\mathbb{G}) \cap (L^\infty(\mathbb{G}) \setminus X) \neq \emptyset$ and one of two things: either X is invariant or $\omega \in M^r(\mathbb{G})$. Therefore we ask the following general question.

Question 5.4. *Let $\mathbb{G}$ be a CQG and $\omega \in M^u(\mathbb{G})$ a contractive idempotent. Do we have that $M_\omega^l(L^\infty(\mathbb{G}))_\perp$ admits a brai if and only if $\mathbb{G}$ is coamenable?*

Acknowledgments

Figure 1: Questions 5.3 and 5.4 in the source, PDF p. 30.

Under the source's correspondence between idempotent states and compact quasi-subgroups, $\epsilon_{\mathbb{G}}^u$ corresponds to

$$N = M_{\epsilon_{\mathbb{G}}^u}^l(L^\infty(\mathbb{G})) = L^\infty(\mathbb{G}).$$

Consequently

$$J^1(N) = N_\perp = \{0\}.$$

The constant net $e_i = 0$ is a bounded right approximate identity for this ideal: the relation $f * e_i \rightarrow f$ is automatic for its sole element $f = 0$.

However,

$$\epsilon_{\mathbb{G}}^u \notin M^r(\mathbb{G}).$$

Indeed, membership in the embedded reduced measure algebra would make the universal counit factor through the reducing morphism $C^u(\mathbb{G}) \rightarrow C^r(\mathbb{G})$. This supplies a reduced counit, which for a compact quantum group is equivalent to coamenability, contradicting the choice of $\mathbb{G}$.

Thus $J^1(N)$ has a brai although $\omega_N = \epsilon_{\mathbb{G}}^u \notin M^r(\mathbb{G})$, disproving Question 5.3. For Question 5.4, the relevant ideal is likewise

$$M_{\epsilon_{\mathbb{G}}^u}^l(L^\infty(\mathbb{G}))_\perp = L^\infty(\mathbb{G})_\perp = \{0\},$$

which has a brai although $\mathbb{G}$ is not coamenable. This disproves the proposed equivalence. □

3 Scope, verification, and novelty

The counterexamples use no pathology beyond the convolution unit. They show that the two universal statements need a nondegeneracy hypothesis. Natural repairs include requiring $N \neq L^\infty(\mathbb{G})$, requiring the ideal to be nonzero, or requiring the multiplier range to be proper. The source's earlier

sufficient theorem uses an intrinsic-group element outside the range, which automatically excludes this example. None of those repaired questions is answered here.

The proof audit checks that the counit is an idempotent state, its multiplier is the identity, the whole algebra is an allowed compact quasi-subgroup, the zero ideal has a brai, and a reduced counit would force coamenability. No computation is relevant.

A bounded search on 11 August 2026 covered the four run indexes, exact question wording, the source title and arXiv id, the published version, the author's thesis, counit/contractive-idempotent combinations, and later citations. No explicit statement of this edge-case counterexample was found. Novelty evidence is provisional.

Human-review recommendation

Check the two conventions on which the edge case rests: the whole algebra is not excluded from the source's compact quasi-subgroups, and the zero ideal is allowed to have the constant-zero brai. Then verify the familiar equivalence between descent of the universal counit and coamenability.

References

- [1] B. Anderson-Sackaney, *On ideals of L^1 -algebras of compact quantum groups*, Int. J. Math. **33** (2022), 2250074; arXiv:2111.13247, Questions 5.3–5.4.
- [2] E. Bédos, G. J. Murphy, and L. Tuset, *Co-amenability of compact quantum groups*, J. Geom. Phys. **40** (2001), 130–153.